AtomS3-Lite

SKU:C124























Description

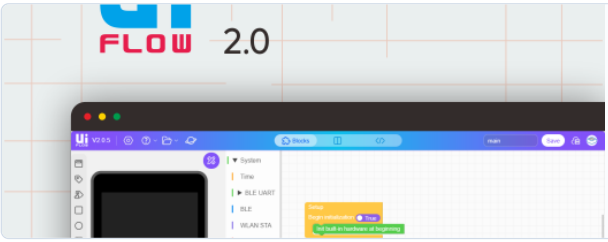
AtomS3-Lite is a programmable controller of the Atom series within the M5Stack development kit lineup, featuring **ESP32-S3** as the main controller and measuring just 24.0 x 24.0 x 9.5mm. It adopts the **ESP32-S3FN8** solution with **8MB** SPI flash, integrated **Wi-Fi**, built-in **3D antenna**, and a **5V to 3.3V** power circuit. The device offers an **IR transmitter LED**, **RGB status indicator**, **button**, and **HY2.0-4P** interface. Power pins and six GPIO female headers are routed out at the bottom for easy expansion. The onboard **USB Type-C** port enables program download and serial communication, and an **M2 screw hole** on the back provides mounting options. This module is ideal for various embedded smart device applications.

Tutorial



Arduino IDE

This tutorial will introduce how to program and control the AtomS3-Lite device using the Arduino IDE.



UiFlow2

This tutorial will introduce how to control the AtomS3-Lite device through the UiFlow2 graphical programming platform.

Features

- Developed based on ESP32-S3FN8
- Built-in USB download function
- Programmable button
- Infrared transmission function
- Expandable pins and interfaces
- Development Platform
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x AtomS3-Lite

Applications

- IoT nodes
- Microcontrollers

Specifications

Specification	Parameter
SoC	ESP32-S3FN8 @ Xtensa® 32-bit LX7 dual-core processor, clock frequency 240MHz
Flash	8MB
RGB	WS2812C-2020
DC-DC	SY8089
Operating Temp	0 ~ 40°C
Power Voltage	5V
Power Method	Type-C
Output Voltage	3.3V
IO Interface ×6	G5/G6/G7/G8/G38/G39
Product Size	24.0 x 24.0 x 9.5mm
Product Weight	5.3g
Package Size	85.0 x 65.0 x 12.2mm
Gross Weight	11.0g

Learn

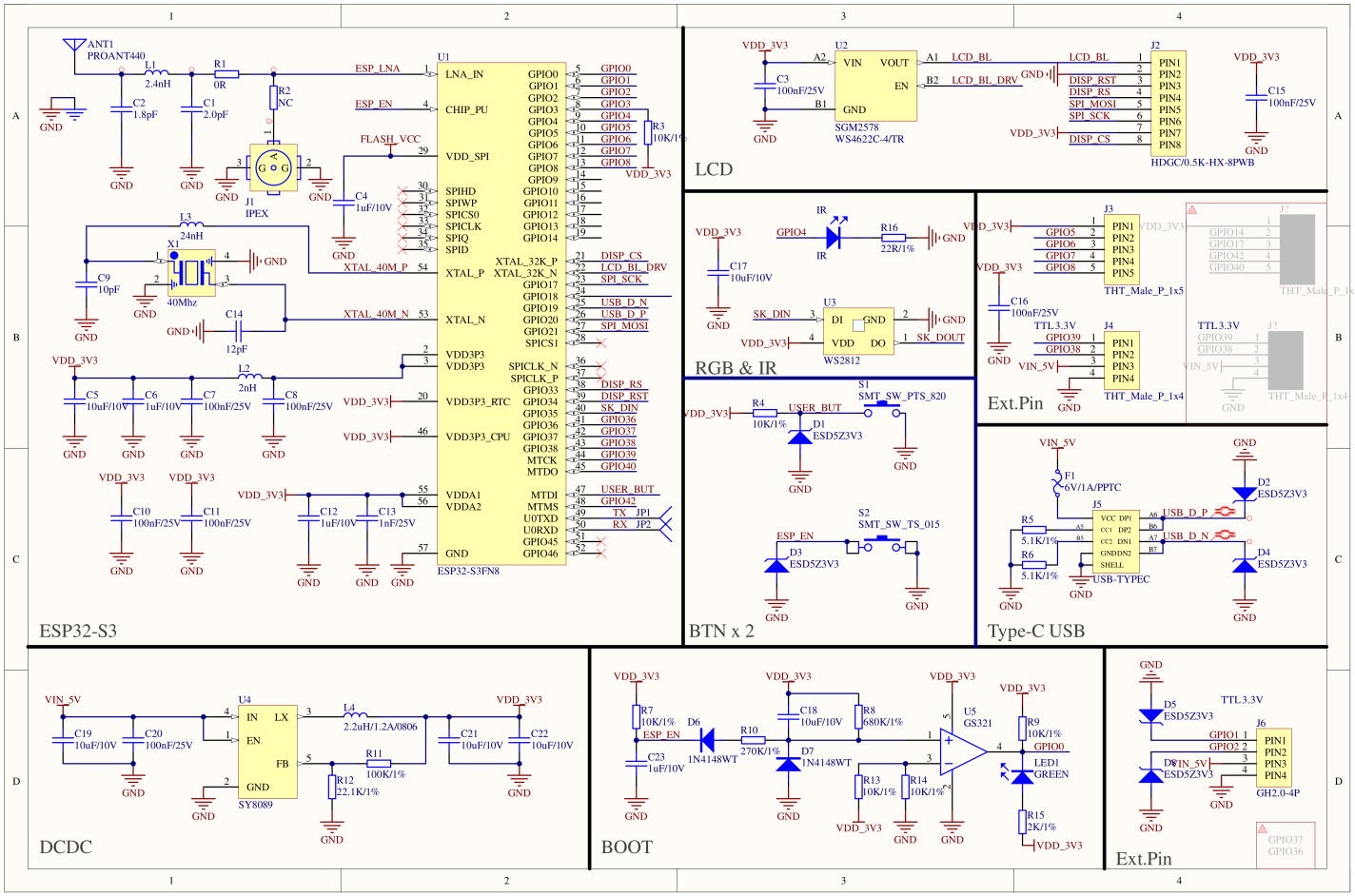
Download Mode

To flash firmware, press and hold the reset button (for about 2 seconds) until the internal green LED lights up, then release the button. After release, the green LED turns off, indicating that the device has entered download mode and is ready for flashing.

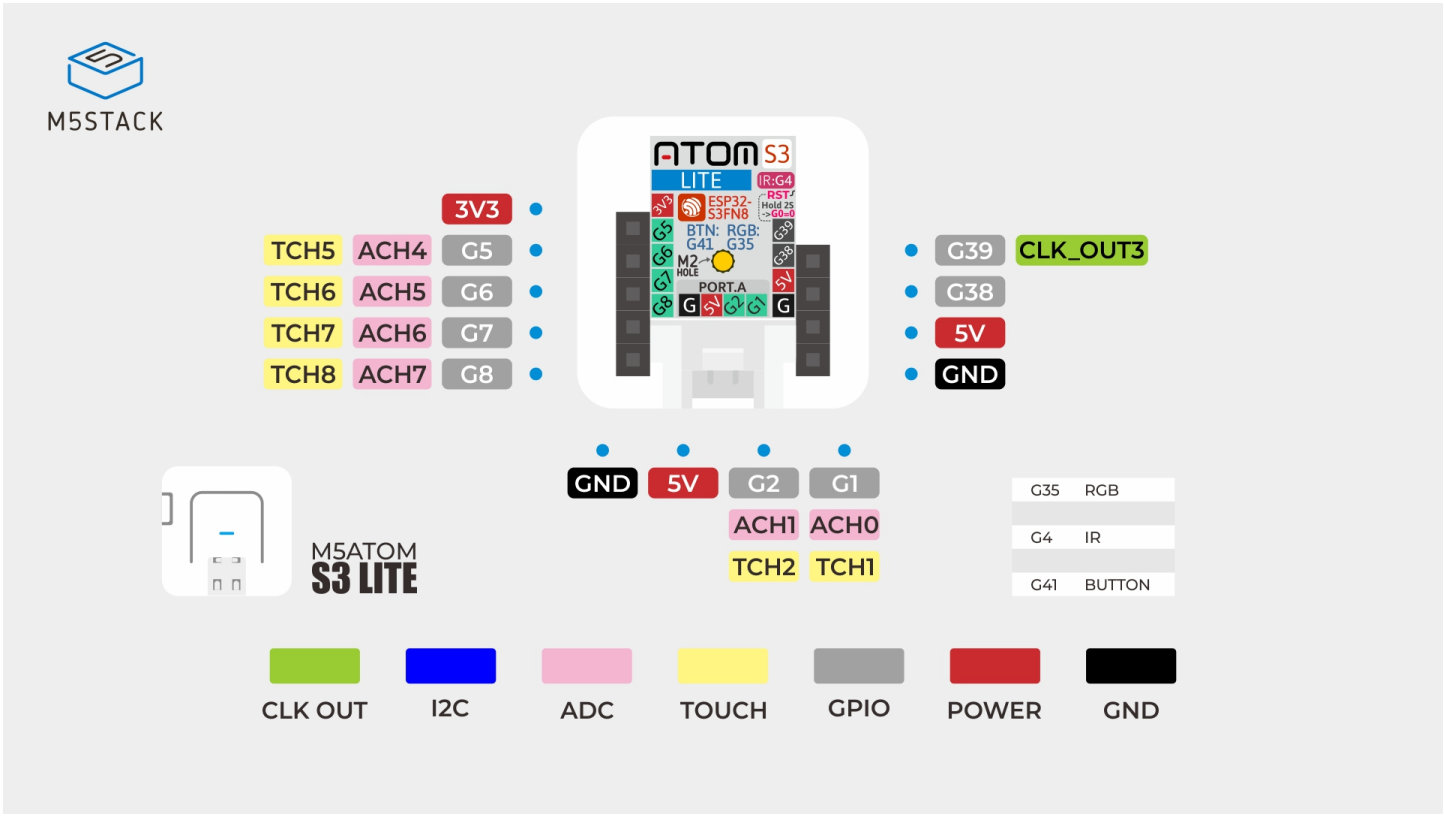


Schematics

- AtomS3-Lite Schematics PDF



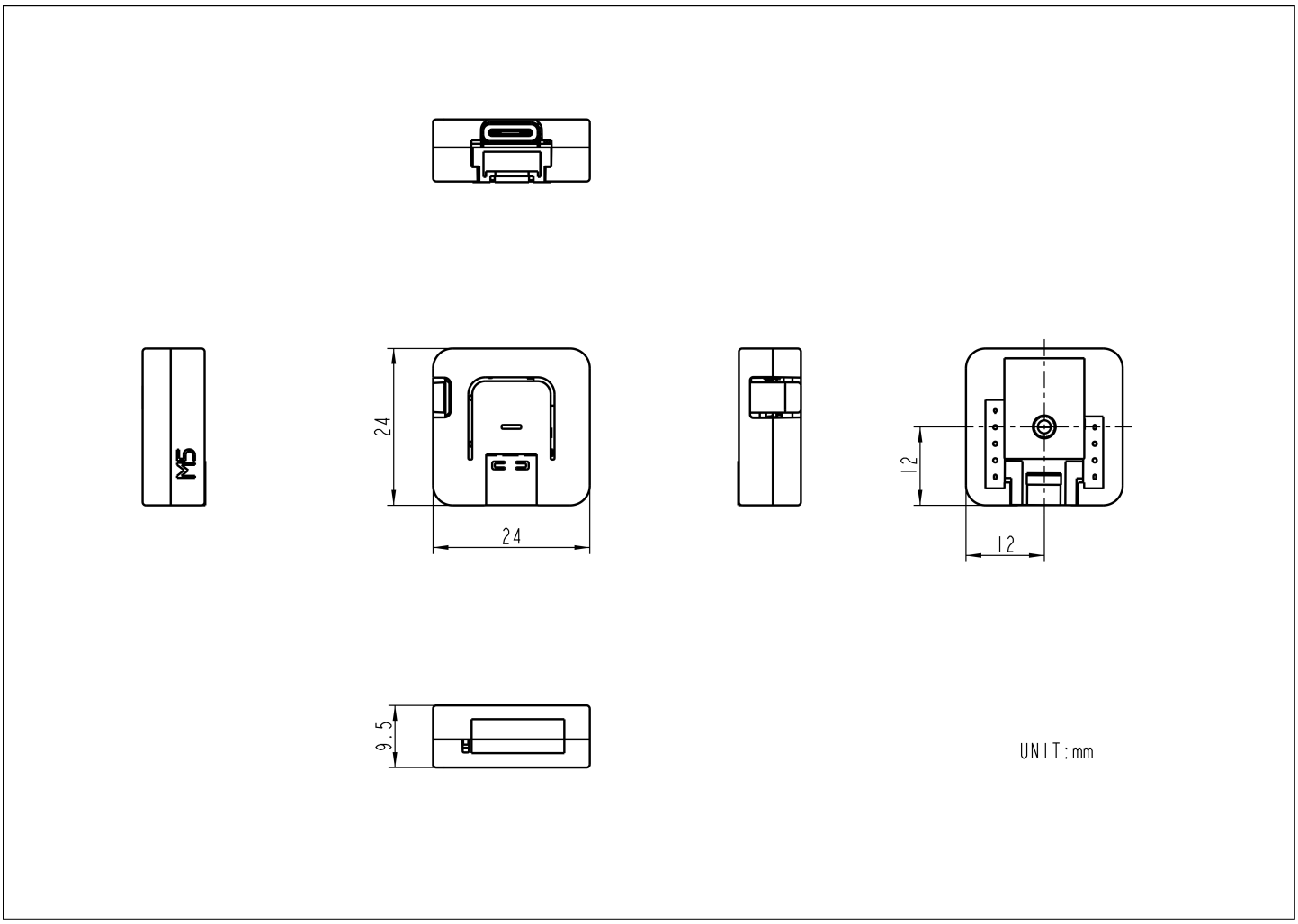
PinMap



HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	G2	G1

Model Size



Structure

- AtomS3-Lite Structure Files

Datasheets

- ESP32-S3
- SY8089

Softwares

Arduino

- AtomS3-Lite Arduino Quick Start
- AtomS3-Lite Arduino Library

UiFlow2

- AtomS3-Lite UiFlow2 Quick Start

PlatformIO

```
[env:m5stack-atoms3]
platform = espressif32
```

```
board = esp32-s3-devkitc-1
framework = arduino
upload_speed = 1500000
monitor_speed = 115200
build_flags =
    -DESP32S3
    -DCORE_DEBUG_LEVEL=5
    -DARDUINO_USB_CDC_ON_BOOT=1
    -DARDUINO_USB_MODE=1
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

Easyloader

Easyloader	Download	Note
AtomS3-Lite Factory Firmware Easyloader	download	/

Video

Product Comparison

To compare information on the Atom series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.